Enabling Real-world Discovery and Innovation v3

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WORLD MODELS – TECH CHASING REALITY

AWS calls World Models "the industry pivot you need to see coming, a whole new world."
Their *5/4/2026* post identifies AMI, World Labs and NVIDIA Cosmos as starters that are not competing but making parallel bets, apparently forgetting that in Jan 2026 Google DeepMind CEO Hassabis announced world models as their primary research goal.

AWS says that the industry pivot you need to see coming is world models that learn as children do, by watching how things work and interact in real life, and this is advocated by Yann LeCun. Turing suggested this over seven decades ago. Research into how infants learn became serious only in the last few decades, and implementing the findings in a computing architecture requires multidisciplinary biomimicry, my area of focus since 1995. The table in the image provides an overview of the current differentiation between world models and real-world models.

	World Models	Biomimetic Real-world Models
Current Status	Initial stages of theoretical research with no roadmap for world modeling tools or products.	- Mature technology with cross-industry history of achievements, including two recently published papers in genomics research. - Supporting analysis of medical AI metadata - Significant innovation and product roadmap.
Purpose	Business: Fill the capability limitations and the value gaps of current AI. Technology: Enable computers to learn like humans.	Business: 1 Enable discovery and learning in highly complex environments. 2 Support innovation and development of users' IP Technology: 1 Enable dynamic categorization of new data and dynamic response to unknown scenarios and events. 2 Capture of human expertise as a computational asset.
Key Method	Develop new theory (AMI – JEPA)	Biomimicry – imitate what works
Input	Pending theory and application	Diverse, siloed and sparse data without cleansing
Output	Pending theory and application	Unseen relationship patterns with the associated real-world evidence
Infrastructure	Presumed significant power and perhaps new chips	Standard infrastructure (pre-AI)
Integration	Presumed with robotics	Stand-alone models or as a component of any platform or ecosystem

March 10, 2026, NABLA posted: Our partners at Advanced Machine Intelligence (AMI) announced a $1.03 billion funding round to accelerate the development of a new class of AI systems known as world models, but what does this mean for healthcare? The post continues: “Clinical environments are dynamic systems involving multiple care teams, fragmented technology infrastructure, evolving patient conditions, and decisions that unfold over time. Supporting that complexity requires AI systems capable of more than generating text, systems that can understand situations, anticipate outcomes, and operate reliably within real-world workflows. This is where world models will profoundly shift healthcare.”

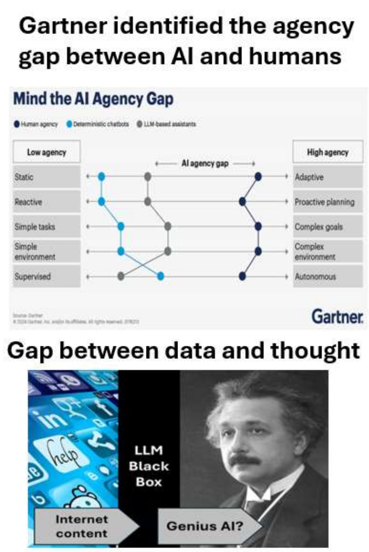
My focus is on discovery and innovation, and in a recent interview Nobel Laureate Jennifer Doudna, the inventor of the Crispr gene-editing technology, was skeptical about AI replacing human effort in the realm of scientific discovery. “I think that innovation is still really in the domain of human beings right now,” she said. “I’m not seeing chatbots coming up with a brand-new idea.”

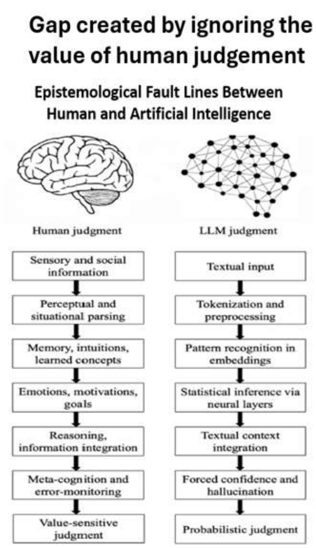
Of course, healthcare is not the only industry where the capability limitations and the value gaps of AI have become evident.

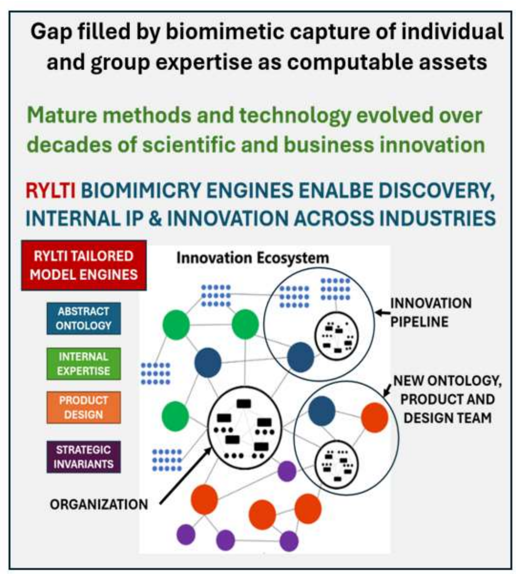
THE AI REAL-WORLD LEARNING GAP

The gap that impedes computational discovery and innovation is multifaceted, as seen in papers by its largest proponents:
“The Illusion of Thinking” – Apple
“The Illusion of Readiness” – Microsoft
“The Abstraction Fallacy” – Google DeepMind
“Why Language Models Hallucinate” – OpenAI
The real-world learning gap is not the topic of these papers, but they elucidate contributing factors.

Key human-computer gaps, and the biomimetic innovation architecture that addresses them.







Innovation requires expertise, and expertise requires learning, but not just acquiring and updating information. It includes understanding, interpretation, and application at the basic level of value. Innovation that delivers significant value involves at a minimum exploration, analysis, and discovery, which are abstract and conceptual thinking processes. Think Einstein and Mendeleev.

Computing can help humans to learn and innovate if the processes are designed to encourage and support thinking, rather that the bizarre goal of replacing human expertise. If “bizarre” strikes you as an extreme adjective, please imagine this scenario: a six-year-old genius who can speed read, memorize, and summarize limitless content; she aces all exams from top schools in business, engineering, and science; would you hire her as an enterprise CEO? If she passed all the medical exams, would you let her care for your health?

The medical aspects of the issue were brought to my attention in 1995 by Dr. David Cohen, the head of intern training at the Albert Einstein College of Medicine. He asked me to design an application to help interns learn differential diagnosis – looking at the relationships across the diverse data to identify as many potential causes as possible for the set of symptoms.

His challenge connected in my mind with a psychological study I read (Mayer 1993) about the differences in problem solution approaches between experts and educated novices. In summary: Comparison of syntactic, semantic, schematic, and strategic differences in problem analysis and solution approaches between recent graduates with advanced degrees and recognized experts in physics, computer science and medicine revealed the following common and distinguishing characteristics of experts:
1. Rapidly and effortlessly recognize issues and anomalies
2. Work with mental models that connect observations and input
3. Manipulate large clusters of information based on context
4. Analyze and plan abstractly and consider many alternatives

The cognitive behavior of professional novices in each of the four above areas was the inverse – looking for the shortest path to familiar solutions.

The novice approach describes LLM computing – homogenization of training data to the mean to produce a prediction.

Using the biomimetic approach illustrated in the image above on the right produced a successful design. It was the beginning of my journey to the current mature methodology and platform.

INTELLIGENCE

The mission of both AMI Labs and RYLTI is to expand computing intelligence to address the value gaps and capability limitations of current technologies. Definitions of intelligence abound, and in technology innovation, the value of the definition is dependent on its support for the objectives of the innovators. What follows is a comparison of the definitions of intelligence by Yann LeCun at AMI, the most prominent advocate of world models, and the earliest builder of real-world models, myself at RYLTI, in the context or our companies’ published business objectives.

AMI (from LeCun's JEPA presentation on YouTube)

Objectives:

- Become the main provider of intelligent systems in all major sectors of the economy
- Build an integrated, modular cognitive architecture around a world model that predicts the consequences of imagined actions and interventions

Intelligence:

- The ability to handle new situations with little or no learning

RYLTI (from my posts)

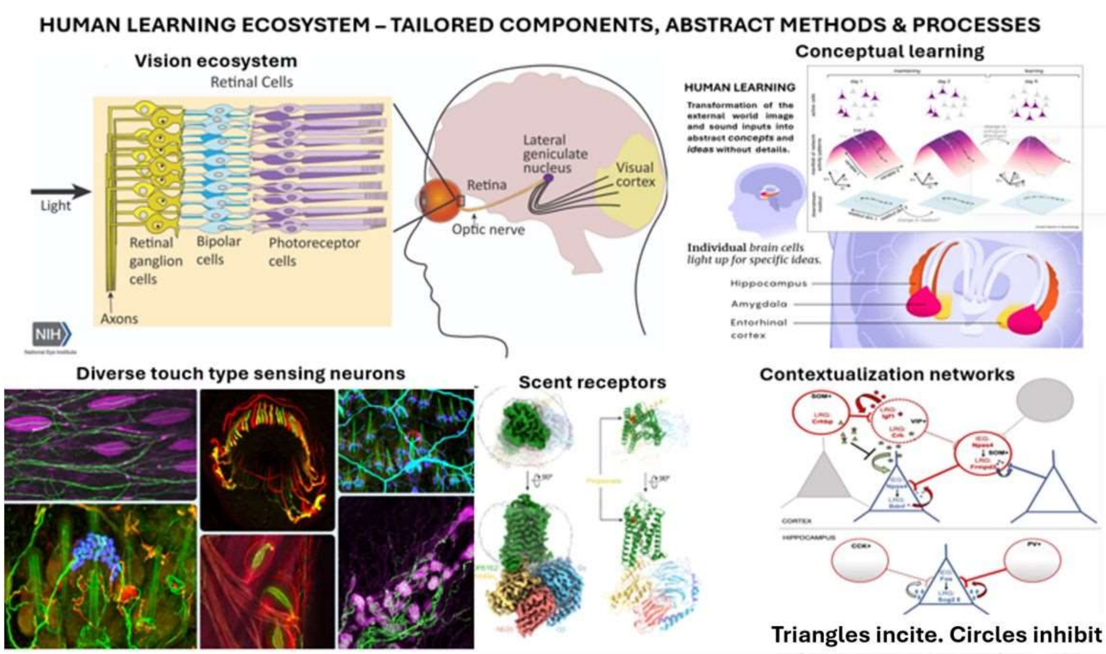
Objectives:

- Enable discovery and learning in highly complex environments and support innovation and development of users' IP
- Capture of human expertise as a computational asset

Intelligence:

- Dynamic learning enabled by sensing, categorization, conceptualization and interpretation of new data and exploring mental models to tailor response to unknown scenarios and events

Clearly there is overlap but the differences are significant. In LeCun's definition of intelligence, we question the value of response without learning. Humans are capable of making quick decisions and some researchers call this prediction, but all organizations value and pay highly for expertise, the product of significant experience and learning.



Modeling expertise requires a significant departure from traditional tools and technical approaches used to predict. My definition of intelligence above is based on the human learning ecosystem, which begins with filtering and categorization at the sensory level (image above), and an expert uses contextual analysis to create a mental model of options and tradeoffs.

That mental model describes an output of a biomimetic real-world model – potentially relevant scenarios and associated evidence.

Simulating the complexity, precision, diversity and adaptability of human learning is clearly beyond our reach, but the Physics of Life report published by the National Academy of Sciences in 2023 provided a real-world alternative – rather than trying to reproduce the mechanisms of life, we con revolutionize computing by discerning principles of living systems and imitating them.

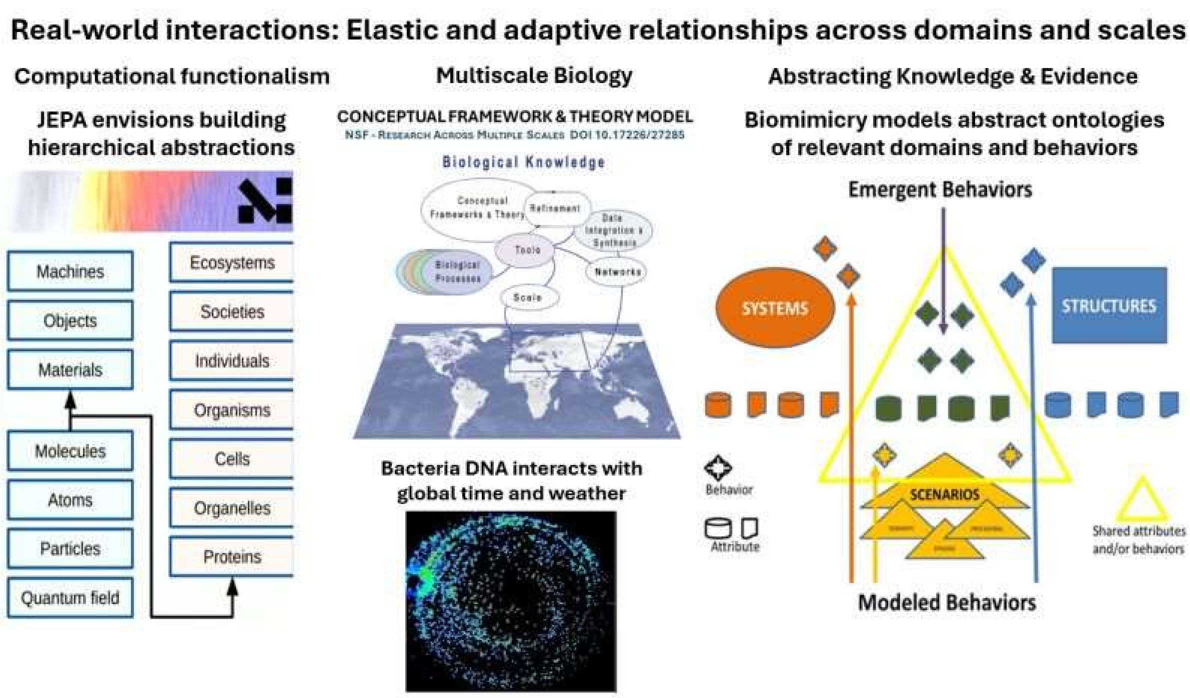
ABSTRACTION

"Science and modeling are all about finding the right representation at the right level of abstraction"
– Yann LeCun, JEPA presentation

I agree, but achieving that goal involves premises that drive the selected methodology, and if the premises do not align with the real world then our models are not scientifically valid. There are implicit premises associated with the two key phrases in Yann's statement.

The first phrase is "finding the right representation" and the key concept is FINDING, which involves exploration, discovery, and learning. The "P" in JEPA is prediction, so the premise is that unknown relevant and valuable abstract representations can be "found" using mathematic methods and the system's knowledge base. How can we abstract what we haven't learned yet?

The second phrase is "right level of abstraction" and the key concept is LEVEL, singular, the premise being that relevant and valuable representations of the world are confined to one abstraction level that is embedded in a hierarchy. Functionalism assumes a linear hierarchy where computation can abstract upwards in the ontology (left image).

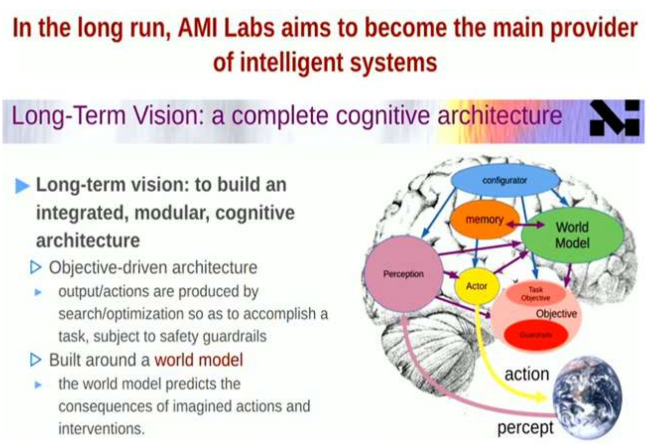


These premises do not align with the real world, composed of innumerable ecosystems, interacting dynamically across scales and domains. The National Science Foundation (NSF) recommends an ecosystem modeling approach across scales and domains including conceptual frameworks and theories (top center image). An example of interaction across scales is cyanobacteria (lower center image). In addition to a cycadean clock to toggle operations between photosynthesis and nitrogen production, gene expressions (illuminated in the lab image) respond to seasonal changes, adding a protective layer outside the membrane in the fall and removing it in the spring.

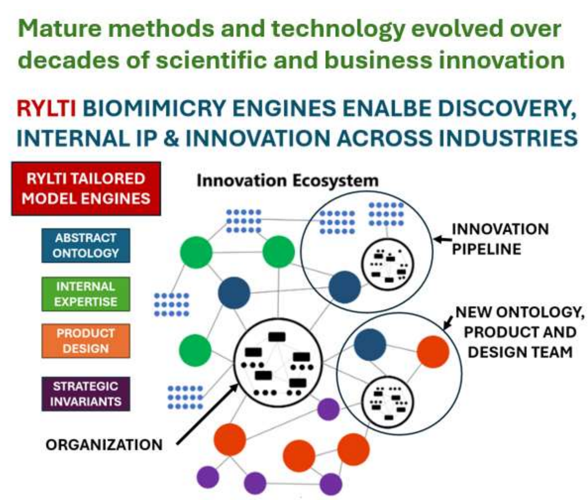
Predictions are useful, but in significant new scenarios validation evidence and judgement need to precede acting on them. As Einstein wrote, “As far as the laws of mathematics refer to reality, they are not certain; as far as they are certain, they do not refer to reality.”

My real-world modeling methodology enables discovery by modeling abstract ontologies of the existing knowledge relevant to a purpose or a problem into an ecosystem of structures, systems, and scenarios (right image above). Rather than making predictions, potentially relevant multiscale and multidimensional relationship scenarios are discovered and output for the analyst to assess their relevance to the analytical objectives. You can find examples in the About section of my LinkedIn profile.

OVERVIEW: AMI JEPA & RYLTI Biomimicry



Source: Yann LeCun JEPA Presentation



The missions and achievements of the two entities are significantly different:

- AMI wants to sell intelligent systems to the world
- RYLTI has been enabling expert users to discover and innovate.
- AMI wants to become the main provider of its technology
- RYLTI is developing global partnerships with diverse entities to create a network of discovery, and to enable innovators to incorporate their internal expertise into their IP and products

Our product pipeline is growing, and we welcome innovation partners.

BIOMIMETIC MODEL GRAPH THEORY (intro)

"What we actually need is a rigorous ontology of computation," concluded a recent paper from Google DeepMind titled The Abstraction Fallacy: Why AI Can Simulate But Not Instantiate Consciousness. The abstract explains: If an artificial system were ever conscious, it would be because of its specific physical constitution, never its syntactic architecture.

Exhaustive modeling of the specific physical constitution of any significant element of the real world is out of reach due to staggering combinatorial complexity, multidimensional and multiscale relationships, and the limits of our understanding.

We can, however, leverage biomimicry to architect learning systems. The computation ontology foundation must be conceptualization, categorization, and interaction relevance. I am applying some biomimetic principles as follows:

Conceptual Categorization

Conceptual categories enable the integration of diverse data by modeling in two phases: an abstract model, or ontology that models the domain being studied independently of the schema of any actual data used, and semantic metadata that assigns a meaning in terms of the ontology to an actual data table or item.

The Model Graph

The ontology, semantic metadata, and actual data combine to form a model graph, containing nodes and links. Useful information is obtained by querying the model graph.

Model Nodes

There are two types of model nodes: concepts, which are declared in the ontology, and objects, which are discovered in the data.

Concepts

There are two types of concepts: those that are declared independently, and those that are part of a classification system, a group of related concepts for which model logic is defined jointly.

Objects

Each object belongs to a class declared in the ontology and has a unique identifier. Each distinct class / identifier pair found in the data represents a single node.

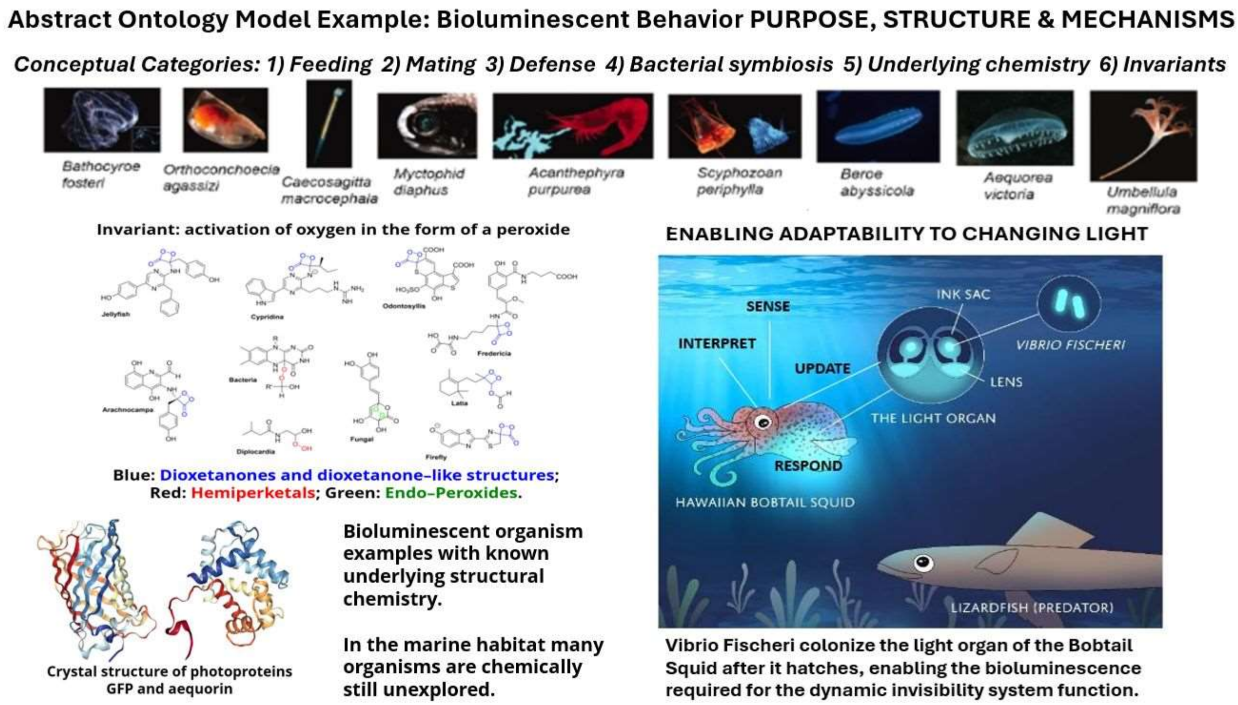
Alternate Identifiers

In some cases, two tables may use different identifier systems to refer to the same object. For example, in most countries a motor vehicle must have a visible identification plate affixed to it with a unique code, and the manufacturer engraves a different Vehicle Identification Number (VIN).

Equivalent Concepts and Objects

Computing equivalence relative to an analysis objective is a significant discovery tool.

Example: Bioluminescent Behavior graph (illustrative image)



Bioluminescence, the natural phenomenon where living organisms produce light through chemical reaction, is beautiful and fascinating. I believe that most of the image is conceptually self-evident (details in research review DOI 10.1002/cbic.202400106).

Enabling Adaptability to Changing Light behavior is an outlier because bioluminescence is used to create invisibility (graphic source: Harvard).
PURPOSE: Hawaiian Bobtail Squid stealth mode while hunting at night
CONCEPTUAL CATEGORY: Defense
STRUCTURE & MECHANISMS: squid hatches with a light sensor above and an underdeveloped light organ below; a luminescent bacteria (Vibrio Fischeri only) colonizes the light organ and stimulates gene expressions that enable full development; when the squid hunts at night, the sensor measures the moonlight/starlight above and opens the light organ cover underneath to expose just enough luminescence to match the above light, making the squid invisible to predators below.

This outlier behavior could not have been discovered using any predictive architecture, regardless of available data and computing power. Princeton researchers performed multiple iterations and integrations of multidisciplinary models. It opened the door to discovery of the microbiome, with significant value for biomedical research. That is an advantage of biomimicry – scaling value without scaling resources.

I agree with DeepMind that we need a computation ontology to model physical constitution related to specific behaviors. I have been working on it for some time. Examples in the About section of my profile.

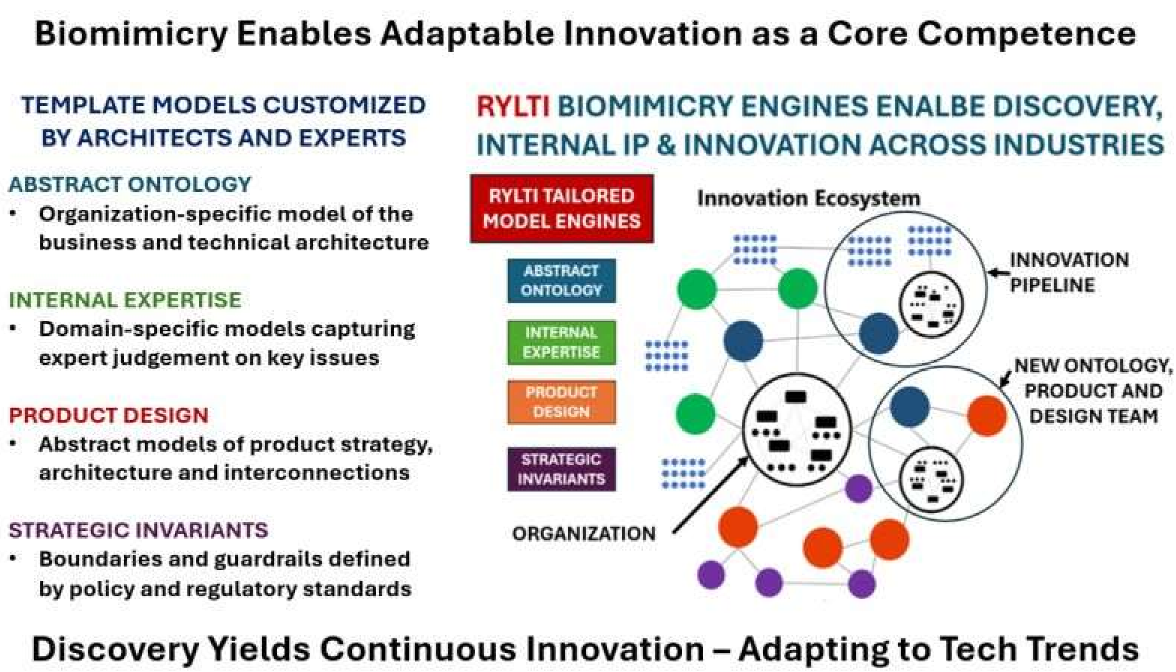
INNOVATION as a CORE COMPETENCY

The rollercoaster of technology trends drives uncertainty and risk for investors, managers, and organizations across industries as novelty speeds into commoditization and obsolescence. The efficient and adaptable creation of value by living systems, from photosynthesis to the human brain, provides principles we can apply to an innovation roadmap that bypasses the rollercoaster.

The roadmap is a transition of innovation from circumstantial opportunism to an organizational core competency. My 30+ years delivering innovation for clients in global enterprise, academia, and government led me to identify two critical success factors to such a transformation:

1. Executives who understand the value of the transformation
2. Methodology and software that leverages the current business and technical architecture

Item #2 (overview in the image) has been developed in the real-world lab of enterprise solution delivery, and there are examples in the About section of my profile. Item #1 is a bit challenging due to the trends rollercoaster and associated hype and FOMO, but the early adopters maintain the competitive moat.



The RYLTI platform is a set of purpose-focused model engines, each with libraries of concepts, objects, links, and rules that are dynamically assembled into models that deliver outputs based on the input parameters. In the image, the bullets on the left are the starting point of the implementation, and the diagram on the right shows the end state of the transformation.

The Starting Point: the template models in each of the engines are customized to the organization by internal technical architects and subject matter experts.

ABSTRACT ONTOLOGY – an organization-specific model that captures the design of the business and technical architectures. Internal architects map the concepts in the ontology to the relevant systems and databases, including sparse datasets, and connect via the REST API.

INTERNAL EXPERTISE – innovation leaders identify the concepts that are relevant to the strategy of the organization. SMEs identify the issues associated with the relevant concepts, including options and tradeoffs, defining domain-specific models that capture expert judgement on key issues.

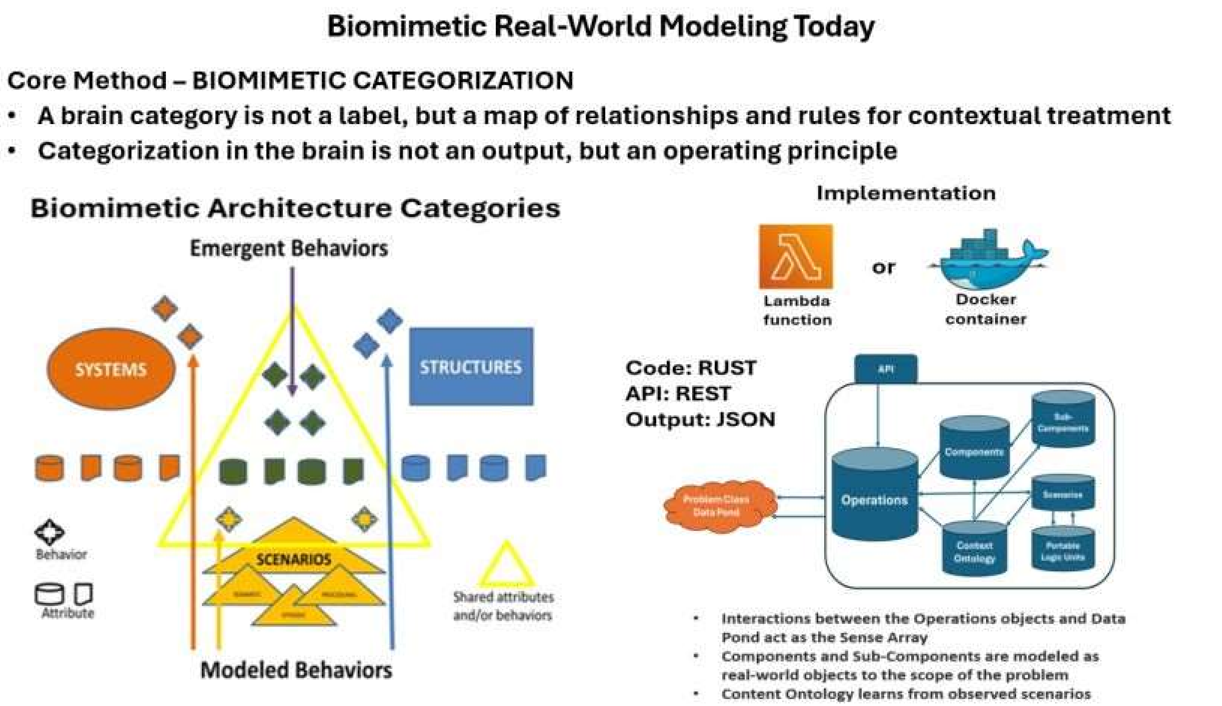
PRODUCT DESIGN – product leaders map the identified concepts to their innovation roadmap and pipeline, building abstract models of product strategy, architecture, and interconnections.

STRATEGIC INVARIANTS – quality and compliance SMEs map relevant design concepts to required boundaries and guardrails defined by policy and regulatory standards.

The Transformation: the implementation must be iterative
- Discrete knowledge domains modeled one at a time across the model ecosystem
- At each stage, exploration processes enable discovery and learning

This is quite different from mainstream technology. We would be happy to do a limited-scope POC to help visualize the value.

THE PIVOT IS HERE
World models are the future, according to AWS, Google DeepMind, AMI, NVIDIA, and others. I agree that world models are the road to more intelligent systems, but the good news is that we do not have to wait.



Biomimetic world models today are elucidating dark data to enable discovery and learning using the architecture illustrated in the image. You can find examples in the About section of my profile.

Additional good news is that it is designed to leverage standard technology and architecture and does not require GPUs or new data centers. 😊

The core method driving discovery and innovation is BIOMIMETIC CATEGORIZATION. A category in the brain is not a label, but a map of relationships and rules for contextual treatment (Nature April 13, 2026, DOI 10.1038/s41583-026-01036-2). Categorization in the brain is not an output, but an operating principle.

Real-world categorization is conceptual. My approach to Conceptual Categorization is introduced in the Biomimetic Model Graph Theory section above.

There are three architectural model categories:
Structures: stable state models that include relevant invariable concepts and objects
Systems: dynamic state models that adapt to changes in the data
Scenarios: stateless models that fulfill an analytical request

This is a high-level overview, of course. The methodology is differentiated from what is currently available, and enables discovery, learning, and adaptability that is out of reach for existing systems.